MORPHISMS OF GRADED RINGS GIVE MORPHISMS ON A PROJ LOCUS

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Allowed standard prerequisites (editorial).

- Sections of twists on standard opens. lemma-proj-sheaves (same chapter, section-level display, not carded). Identifies O_X(n) restricted to D_+(f) with the degree-n part of A_f as an A_(f)-module.

Lemma 0.1. *Let A, B be two graded rings. Set $X = Proj(A)$ and $Y = Proj(B)$. Let $\psi : A \to B$ be a graded ring map. Set*

$$U(\psi) = \bigcup\nolimits_{f \in A_+ \text{ homogeneous}} D_+(\psi(f)) \subset Y.$$

Then there is a canonical morphism of schemes

$$r_\psi : U(\psi) \longrightarrow X$$

and a map of $\mathbf{Z}$-graded $\mathcal{O}_{U(\psi)}$-algebras

$$\theta = \theta_\psi : r_\psi^* \left(\bigoplus\nolimits_{d \in \mathbf{Z}} \mathcal{O}_X(d) \right) \longrightarrow \bigoplus\nolimits_{d \in \mathbf{Z}} \mathcal{O}_{U(\psi)}(d).$$

The triple $(U(\psi), r_\psi, \theta)$ is characterized by the following properties:

(1) *For every $d \geq 0$ the diagram*

$$\begin{array}{ccccc} A_d & & \xrightarrow{\qquad\qquad\psi\qquad\qquad} & & B_d \\ \downarrow & & & & \downarrow \\ \Gamma(X, \mathcal{O}_X(d)) & \xrightarrow{\theta} & \Gamma(U(\psi), \mathcal{O}_Y(d)) & \longleftarrow & \Gamma(Y, \mathcal{O}_Y(d)) \end{array}$$

is commutative.

(2) *For any $f \in A_+$ homogeneous we have $r_\psi^{-1}(D_+(f)) = D_+(\psi(f))$ and the restriction of r_ψ to $D_+(\psi(f))$ corresponds to the ring map $A_{(f)} \to B_{(\psi(f))}$ induced by ψ.*

Proof. Clearly condition (2) uniquely determines the morphism of schemes and the open subset $U(\psi)$. Pick $f \in A_d$ with $d \geq 1$. Note that $\mathcal{O}_X(n)|_{D_+(f)}$ corresponds to the $A_{(f)}$-module $(A_f)_n$ and that $\mathcal{O}_Y(n)|_{D_+(\psi(f))}$ corresponds to the $B_{(\psi(f))}$-module $(B_{\psi(f)})_n$. In other words θ when restricted to $D_+(\psi(f))$ corresponds to a map of $\mathbf{Z}$-graded $B_{(\psi(f))}$-algebras

$$A_f \otimes_{A_{(f)}} B_{(\psi(f))} \longrightarrow B_{\psi(f)}$$

Condition (1) determines the images of all elements of A. Since f is an invertible element which is mapped to $\psi(f)$ we see that $1/f^m$ is mapped to $1/\psi(f)^m$. It easily follows from this that θ is uniquely determined, namely it is given by the rule

$$a/f^m \otimes b/\psi(f)^e \longmapsto \psi(a)b/\psi(f)^{m+e}.$$

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To show existence we remark that the proof of uniqueness above gave a well defined prescription for the morphism r and the map θ when restricted to every standard open of the form $D_+(\psi(f)) \subset U(\psi)$ into $D_+(f)$. Call these r_f and θ_f. Hence we only need to verify that if $D_+(f) \subset D_+(g)$ for some $f, g \in A_+$ homogeneous, then the restriction of r_g to $D_+(\psi(f))$ matches r_f. This is clear from the formulas given for r and θ above. $\square$